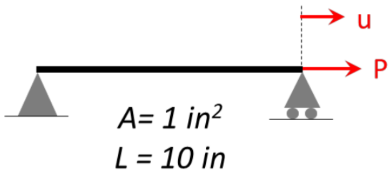
Problem # 3

3. Incremental Iterative Solution: Consider a truss element under axial loading as shown. The material is modeled using Menegotto-Pinto model with the following parameters: $\sigma_y = 60\ ksi, E = 29000\ ksi, b = 0.05, R_0 = 20, R_1 = 0.925, R_2 = 0.15, a_1 = a_2 = 0.$



Write a Matlab code to perform incremental iterative process for quasi-static loadings as detailed below. Use the Menegotto-Pinto material model code provided in the class. Return your commented Matlab code with your solution.

a. Perform a quasi-static monotonic pushover analysis from P = 0 to P = 200 kips using Newton Raphson method. Plot the pushover curve (i.e., force vs. displacement). Plot the iteration points on top of the pushover curve.

   Note: a good force step size is 3-5% of the yield force.

b. Repeat part (a) with modified Newton algorithm using the initial elastic stiffness. Plot the convergence criterion vs. number of iterations and compare for the two methods. State your observations.

c. Perform a quasi-static cyclic pushover analysis for P = [0, 100, -100, 200, 0] kips. Plot the cyclic pushover curve. If convergence cannot be achieved, try to investigate the underlying reason(s) and solve the issues. State the reason(s) and the solution you developed.

Part a and b: Observations

Newton-Raphson: The maximum number of iterations in any single load step is only 6, resulting in a total of 234 iterations across the entire pushover analysis. This method updates the global tangent stiffness at each iteration, allowing the algorithm to rapidly find equilibrium in very few iterations, even after significant yielding.

Modified Newton-Raphson: The maximum number of iterations in any single load step is 374, resulting in a total of 26,430 iterations across the entire pushover analysis to complete the identical load history. This method uses the same initial global tangent stiffness throughout the analysis.

Pre-Yield: The structural response is perfectly elastic, so $K_{initial}$ matches the true tangent stiffness, and convergence is achieved in very few steps.

Post-Yield: The member yields, and the true tangent stiffness drops to just 5% of its elastic value (as b=0.5). Since it continues to use $K_{initial}$, it under predicts the necessary displacement correction in each iteration. This leads to a very slow convergence rate.

Part c: Observations, Underlying Reasons, and Solution:

There is convergence issue at each reversal: When the cyclic load path is run with Newton-Raphson, convergence fails at the load steps right after each reversal (each peak). The solver hits the maximum iterations and prints warnings such as:

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Warning: Reached max iterations at load P = 98.0 using Full-NR
Warning: Reached max iterations at load P = 96.0 using Full-NR
Warning: Reached max iterations at load P = 94.0 using Full-NR
….
Warning: Reached max iterations at load P = -98.0 using Full-NR
Warning: Reached max iterations at load P = -96.0 using Full-NR
Warning: Reached max iterations at load P = -94.0 using Full-NR
….
Warning: Reached max iterations at load P = 198.0 using Full-NR
Warning: Reached max iterations at load P = 196.0 using Full-NR
Warning: Reached max iterations at load P = 194.0 using Full-NR
….

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Reason: This is due to the slope discontinuity (the kink) at every reversal. At a converged peak (say P = 100 kips) the material is on the hardening branch, so the stored tangent modulus is Et ≈ bE = 0.05 × 29000 = 1450 ksi, and the structural tangent stiffness is K_T = A·Et/L = 145 kip/in, which is small. The next load step asks for slightly less force, but the displacement correction $\boldsymbol{\delta}$u = ψ / K_T is large. This large correction moves the displacement far past the equilibrium point, across the kink and onto the steep elastic-unloading branch. On the next iteration it overshoots back. The trial strain increment keeps flipping sign, so the material keeps jumping between the loading and unloading branches and never settles within the tolerance in the allowed number of iterations.

Solution: To solve this issue, for the first iteration of every load step we use elastic stiffness,and the consistent tangent for all later iterations:

$K_T = K_{elastic}$ if idx == 0 else (A * state.Et) / L

The elastic stiffness being large (2900 ksi), the first correction $\boldsymbol{\delta}$u = ψ / $K_{elastic}$ is small and lands on the correct (unloading) branch instead of jumping across the discontinuity. After that one small step, the consistent tangent is used which converges quickly.

With this solution, all cyclic load steps converged (at most 7 iterations per step) and the correct hysteresis loop was obtained.

There were no such issues during part a and b because during monotonic loading there were no reversals.